Effect of Eosin Blue Dye on Potentiostatically deposited ZnO Thin Films for DSSC Applications

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***Abstract*— Zinc oxide (ZnO) and zinc oxide/eosin blue (ZnO/EB) thin films were potentiostatically electrodeposited onto fluorine doped tin oxide (FTO) substrate. The influence of eosin blue dye on the structural, morphological and optical properties was investigated with the help of X-ray diffraction (XRD), Field emission scanning electron microscopy (FE-SEM), micro Raman and photoluminescence (PL) analysis. XRD patterns reveal that the deposited thin films are a hexagonal wurtzite structure grown along with highly oriented the (101) plane, and the ZnO/EB film exhibits a good crystallinity compared to pure ZnO. ZnO and ZnO/EB thin films show a flower and densely packed nanowall with porous structure, respectively. In micro Raman spectra, the presence of high intense E_2 (high) mode represents the nanowall possesses a better crystalline quality than that of the flower structure. PL spectra reveal that the nanowall structure has an intensive near band edge emission with less deep level emission related to the atomic defects. Current density-voltage (J-V) curves are recorded for the flower and nanowall structures, and their parameters such as open-circuit voltage (V_{oc}), short-circuit current density (J_{sc}) and efficiency (η) is calculated. Nanowall structured thin film based DSSC exhibits higher V_{oc}, J_{sc} and η than the flower like structure.**

Keywords: DSSC; nanowall; Zinc oxide; J-V curve

I. INTRODUCTION

Solar energy is the best key fossil fuels for clean energy generation without any affects of environmental [1]. This energy can be effectively harvested by solar cells. Among the different generations of the solar cells, the third generation dye sensitized solar cells (DSSCs) have been growing for last two decades due to their simple fabrication, low production cast, easy accessibility and environmental friendly [2]. DSSCs have higher indoor efficiency than the conventional solar cells [3]. In general, the high efficiency has been achieved in DSSC made up of titanium oxide (TiO_2) nanoparticles coated FTO as a photoanode and platinum coated FTO as a counter electrode and iodine redox couple as an electrolyte. Furthermore increasing the efficiency of DSSC beyond the current limit is a difficult due to their low electron mobility in photoanode. Therefore, zinc oxide (ZnO) can be an alternative semiconducting material to TiO_2, because which has a high exciton binding energy (60 meV) with band gap energy (3.37 eV). ZnO is an n-type and II-IV group material with more than three order of electron mobility compared to that of the TiO_2 [4]. The efficiency of the DSSC depends on crystallinity, morphology and optical absorption of the nanostructures. They have been prepared by using physico-chemical routs such as chemical vapor deposition [5], thermal evaporation [6], electron beam evaporation [7], hydrothermal technique [8], sol-gel [9] and electrodeposition [10]. Among the aforementioned methods, the electrodeposition is one of the best methods due to easy preparation, low cost, easy controllable of film thickness as well as morphology, which has been used to prepare the nanostructured thin films such as one dimensional (nanospikes and nanopillar [11], nanorods [12], nanowires [13]), two dimensional (nanosheet [14] and nanowall [15]) and three dimensional (microsphere particle [16]). Since, DSSC's efficiency almost depends on morphology of the photoanode [17]. In especially, the short circuit current density is mainly related to the nanostructures and their crystallinity. In that context, most of the research groups concentrate the preparation of nanoparticles which offer a large surface area for more dye loading [18]. Eventhough, they give more surface area, the injected electron mobility is very low due to the presence of large number of grain boundary in the nanoparticle photoanode, which leads to a recombination between the back transfer electrons and the oxidized dye molecules [18]. To improve the electron mobility and reduce the recombination, one dimensional (1D) nanostructure has been synthesized, which offer a direct electron pathway. But, 1D nanostructure has low surface area with direct recombination between collected electron on the FTO substrate and the oxidized species in electrolyte [19]. In order to increase the surface area as well as the electron mobility, Yoshida et al have synthesized a porous like structure by addition of the eosin yellow organic dyes in deposition solution [20].

In the present work, we attempt to prepare the porous structure by using eosin blue dye. The influence of dye on the prepared thin film were studied by means of XRD, FE-SEM, micro Raman and PL analysis to analysis the structural, morphological, vibrational and

optical properties. The light harvesting efficiency of the constructed DSSCs was characterized by J-V curve.

II. EXPERIMENTAL

2.1. Photoanode preparation

Zinc oxide and zinc oxide/eosin blue thin films were electrochemically prepared by a conventional three electrode system. Prior to thin film deposition, fluorine doped tin oxide (FTO) substrate was rinsed with acetone and deionized water, which was used as a working electrode. Platinum (Pt) and Ag/AgCl were used as a counter electrode and a reference electrode, respectively. A deposition solution contained mixture of 50 mM zinc nitrate hexahydrate, 0.1 M potassium chloride and 5 mM hexamethylenetetramine. The thin film deposition was potentiostatically performed by EG&G Princeton Applied Research potential/galvanostate instrument 362 at a constant potential of -1.2 V at a bath temperature of 80˚C for 30 minutes. In order to study the influence of eosin blue dye, 50 µM of eosin blue dye was added in the deposition solution. After the film deposition, the deposited ZnO/eosin blue film was rinsed with deionized water. Subsequently, the film was immersed into a KOH alkaline solution to remove the eosin blue dye from the deposited film for 24 hours. Thereafter, the films were annealed at 450˚C for an hour at air atmosphere. Finally the films were subjected to characterizations.

2.2. DSSCs construction

The deposited flower and nanowall structured thin films were dipped into a 0.3 mM of ethanolic N719 dye solution for 24 hours. The Pt electrodes were prepared by a sol-gel spin coating technique. The spin coating solution consisted of 5 mM chloroplatinic acid in isopropyl alcohol. The Pt was coated onto FTO substrate, and finally the Pt electrodes were annealed at 385°C for one hour. The area of the cells was fixed about 0.25 cm^2. The dye loaded photoanode and the Pt electrode were sandwiched between electrolyte consisting of 0.6 M 1-methyl-3-propylimidazolium iodide, 0.05 M iodide, 0.1 M lithium iodide and 0.5 M tert-butylpyridine in acetonitrile.

2.3. Characterizations

Crystal structure of the thin films was analysized by using X'Pert PRO PAN analytical powder X-ray Diffraction meter. The morphologies of the ZnO and ZnO/EB films were observed using filed emission scanning electron microscopy (FE-SEM, FEG Quanta 250). Vibrational mode and crystal structure were analysized by micro Raman spectroscopy (Princeton instrument Acton sp 2500). The luminescence properties and crystal quality of the thin films were studied by Varian Cary Eclipse photoluminescence spectroscopy. Current density-voltage curves were measured by Keithley 2400 under the light illumination of 100 mW using Xenon lamp.

III. RESULTS AND DISCUSSION

3.1. Growth mechanism

Figure In the electrochemical deposition method, the growth of ZnO nanostructrues formation are taken place by electrochemical reduction process in the aqueous solution containing $Zn(NO_3)_2$ (zinc nitrate) and $(CH_2)_6N_4$ (hexamethylenetetramine). The $Zn(NO_3)_2$ salt is dissociated into a Zn^{2+} (zinc) and NO_3^- (nitrate) ion in deionized water (H_2O) (equ.1). Further the NO_3^- ion electrochemically reduced with H_2O molecule forms a NO_2^- (nitrite) and OH^- (hydroxide) (equ.2) [15]. $(CH_2)_6N_4$ thermally decomposed with H_2O gives a NH_3 (ammonia) and HCHO (formaldehyde) (equ.3). Subsequently, the NH_3 reacted with H_2O gives a NH_4^+ (ammonium) and OH^- (equ.4) [21]. Finally, the Zn^{2+} electrochemically reduced with OH^- forms a $Zn(OH)_2$ (zinc hydroxide). The $Zn(OH)_2$ is thermally not stable, so that which dehydrates into ZnO and H_2O (equ.5). After the addition of eosin blue (EB) dye in the deposition solution, there is a possible to form the $(Zn^{2+})_2$ (EY^{2-}) complex formation due to reduction of Zn^{2+} and EB^{2-} ions in electrochemical method (equ.6). In thermodynamically, the presence of carboxylic acid group in the EB dye molecules leads to absorb on surface of the ZnO thin films [19].

$$Zn(NO_3)_2\,(s) \longrightarrow Zn^{2+}\,(aq) + 2NO_3^-\,(aq) \;\ldots\ldots (1)$$

$$NO_3^- + H_2O + 2e^- \longrightarrow NO_2^- + 2OH^- \;\ldots\ldots\ldots (2)$$

$$(CH_2)_6N_4 + 6H_2 \longrightarrow 4NH_3 + 6HCHO \;\ldots\ldots (3)$$

$$NH_3 + H_2O \longrightarrow NH_4^+ + OH^- \;\ldots\ldots\ldots\ldots (4)$$

$$Zn^{2+} + 2OH^- \longrightarrow Zn(OH)_2(s) \longrightarrow ZnO(s) + H_2O \;\ldots\ldots (5)$$

$$2Zn^{2+} + EB^{2-} \longrightarrow (Zn^{2+})_2\,(EY^{2-}) \;\ldots\ldots\ldots\ldots (6)$$

3.2. X-ray diffraction analysis

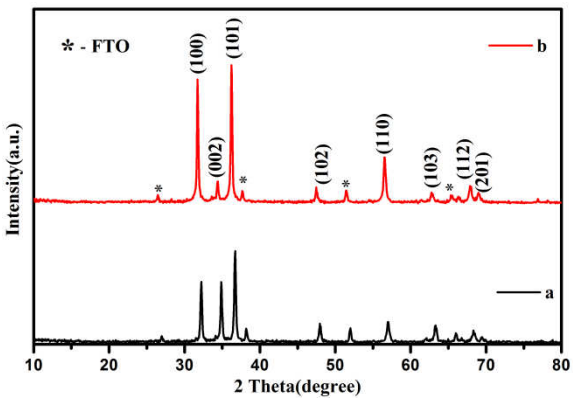

Fig. 1. XRD patterns of (a) ZnO and (b) ZnO/EB thin film.

X-ray diffraction (XRD) analysis was performed to investigate the crystal structure of the deposited thin films. Fig. 1 shows XRD patterns of the ZnO and

ZnO/EB thin films. All diffraction peaks are well indexed and good in accordance with JCPDS card no. 36-1451. The diffraction (100), (002) and (101) peaks indicate that the deposited thin films are hexagonal wurtzite structure. The predominant (101) peak reveals the crystalline growing along their preferential growth orientation in the ZnO film.

When 50 µM of the EB dye is added in the deposition solution, the increased intensity of the (101) peak indicates that crystallinity of the ZnO/EB film is significantly increased compared with the ZnO film [10]. The intensity of (101) peak in the ZnO/EB film is increased with decreasing (002) peak due to their highly preferential growth orientation of the crystalline compared to the ZnO film. The intensity of all other peaks is increased except (103) peak, which is indicated that the growth of crystalline could be parallel to (101) peak. Crystallite size of the films is estimated by using Scherrer equation (7) [22].

$$D = \frac{0.9\lambda}{\beta\cos(\theta)} \text{.....}(7)$$

where λ is the wavelength of the X-ray, β is the full width at half maximum of the diffraction peak and θ is the diffraction angle of the peak. The estimated crystallite size of the ZnO and ZnO/EB thin films is 34.03 and 42.48 nm, respectively. From these XRD study, one can conclude that the crystallinity of the ZnO/EB film is increased compared with the ZnO film. The increased crystallite size of the film decreases the grain boundary, which favoures to increase the electron mobility.

Table 1. Crystallite size, dislocation density and micro strain.

Sample code	Crystallite size (D) nm	Dislocation density (δ) X 10^{14} lines/m^2	Micro strain (ε) X 10^{-3} $line^{-2}$ m^{-4}
a	34.03	8.63	1.35
b	42.48	5.54	1.06

Dislocation density (δ) of the ZnO and ZnO/EB film is evaluated from crystallite size using the equation (8).

$$\delta = \frac{1}{D^2} \cdots\cdots\cdots\cdots\cdots\cdots\cdots (8)$$

The micro strain of the deposited films is calculated using the equation (9).

$$\varepsilon = \frac{\beta\sin\theta}{4} \cdots\cdots\cdots\cdots\cdots\cdots\cdots (3)$$

As shown in table 1, the micro strain of the ZnO/EB film is decreased compared to the ZnO film due to decrease the defects in lattice site. The dislocation density is found to decrease due to release of strain in the ZnO/EB films.

3.3. FE-SEM analysis

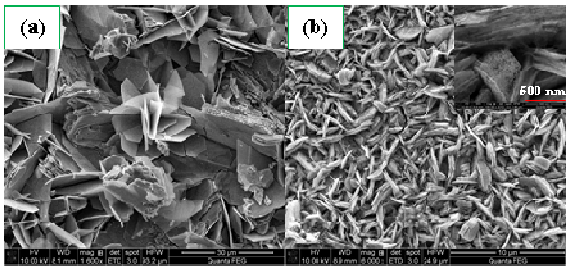

Fig. 2. FE-SEM image of (a) ZnO and (b) ZnO/EB thin film.

FE-SEM image displays a flower like structure for the ZnO thin film as shown in Fig. 1(a), which consists of more number of the sheets. The sheets are very soft. The size of the flowers varies about 22.3-31.6 µm. The length of the sheets varies in the range between 15.0 and 19.9 µm. The formation of two dimensional nanostructures in electrodeposition method depends on the formation of $Zn(OH)_2$ and their dehydrate. As the formation of $Zn(OH)_2$ is greater than dehydrate, the two dimensional nanostructures are formed [15]. When 50 µM of the EB dye is added in the deposition solution, morphology of the prepared film is shown as a nanowall structures which are densely cover on the surface of FTO substrate as shown Fig. 1(b). The length and thickness of the nanowall are about 2.2-2.5 µm and 129-402 nm, respectively. The inset Fig. 2(b) shows a porous in the nanowall structures, which is due to absorption of the dye molecule on surface of the ZnO. The size of the porous varies in the range of 12.5-26.7 nm. The porous structure gives the large surface area for more dye loading, and the electrolyte can easily penetrate into the nanowall structure to regenerate the oxidized dye molecules.

3.4. Raman spectra analysis

Raman spectroscopy is one of the best tools to investigate the vibrational modes, atomic defects and crystal structures of the films. Fig. 3 shows Raman spectra of the flower and nanowall structured films. Raman peaks at 330, 437 and 582 cm^{-1} are related to the E_{2L}-E_{2H}, E_2 (high), and E_1 (LO) mode, respectively [23]. The peak is appeared at 330 cm^{-1} due to the second order non-polar mode of Raman scattering. The peak is observed at 437 cm^{-1} which is ascribed to E_2 (high) mode due to the oxygen vibration [24]. The presence of E_2 (high) mode reveals that the deposited films are hexagonal wurtzite structure. The increased intensity of E_2 (high) mode indicates that the nanowall has a better crystallinity than the flower structured film, which is well consent with XRD results. Raman peak is found to be at 582 cm^{-1} owing to the point defects either oxygen vacancies or Zn interstitials. The Raman peak is appeared in the range of 1050-1200 cm^{-1} owing to the multi phonon process [24]. From the Raman spectra,

we could confirm that the nanowall structure has a high crystal quality compared with the flower structure.

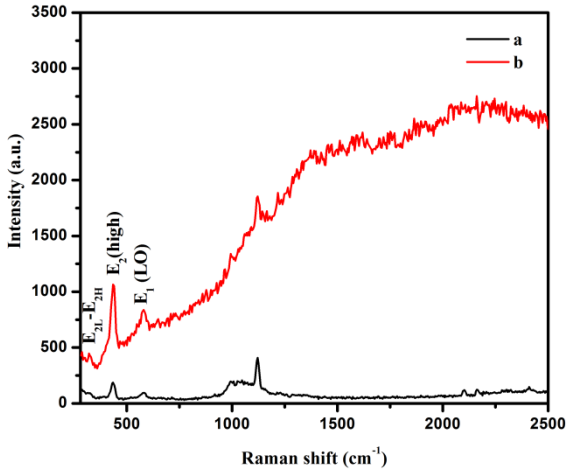

Fig. 3. Micro Raman spectra of (a) flower and (b) nanowall structured thin films.

3.5. PL Spectra analysis

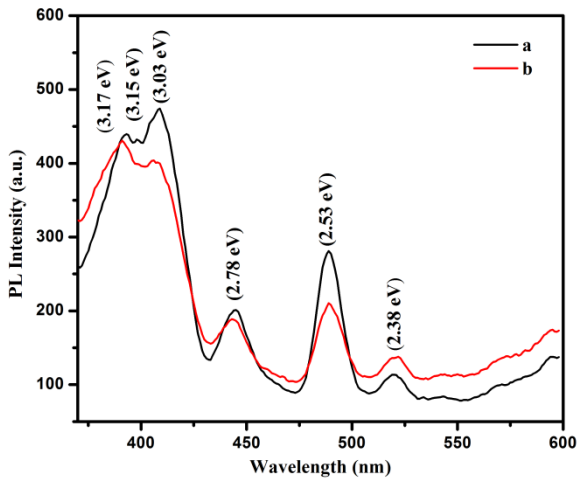

Fig. 4. PL spectra of (a) flower and (b) nanowall structured thin film.

Photoluminescence (PL) spectra were carried out to study the emission properties, atomic defects and crystal quality of the deposited thin films and are shown in Fig. 4. The UV emission peak is appeared at 390 nm (3.17 eV), which is ascribed to a near band edge emission owing to the free exciton recombination by exciton- exciton collision in the nanowall film [25]. The red shift of the UV emission peak from 390 to 393 nm in the flower is due to the presence of native defects such as zinc vacancy, zinc interstitial and oxygen vacancy compared with the nanowall structure. The visible emission peaks at 408, 445, 499 and 520 nm are assigned to a deep level emission. The strong violet emission peak is observed at 408 nm (3.03 eV) due to the energy difference between the conduction band and the zinc vacancy [26]. The blue emission peak is appeared at 445 nm (2.79 eV) owing to the interstitial zinc defects [27]. The green emission peaks are appeared at 491 nm (2.53 eV) and 520 nm (2.38 eV), which are assigned to the energy different between either the zinc interstitial or oxygen vacancy and the valence band [28]. The zinc interstitial or oxygen vacancy is corroborated through the presence of E_1 (LO) mode for both nanostructures in Raman spectra. The crystal quality of the films is estimated by the intensity ratio of UV to visible ($I_{UV/Vis}$) [28]. The UV/Vis ratio of nanoflower and nanowall is found to be 0.92 and 1.06, respectively. From the PL spectra, it is clearly observed that the crystal quality of the nanowall is higher than that of the flower, which is well concurrent with the presence of high intense (101) peak and E_2 (high) mode of the nanowall in XRD pattern and Raman spectra, respectively.

3.6. J-V Characterization

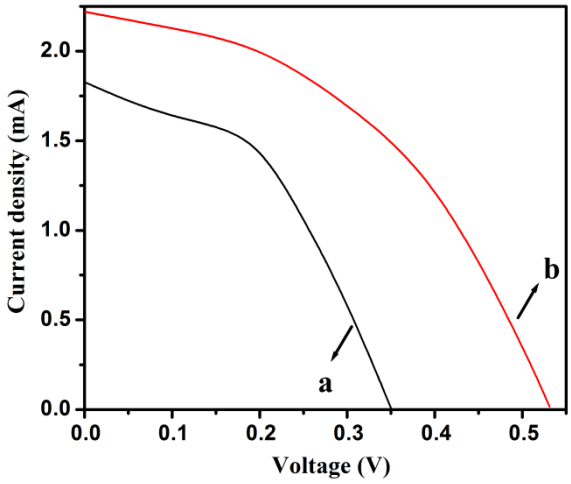

Fig. 5. J-V curve of DSSC based on (a) flower and (b) nanowall structures.

Current density-voltage (J-V) curves were carried out for DSSC based on the flower and nanowall structure to find out the light harvesting ability and are shown in Fig. 5. The calculated parameters are summarized in table 2. DSSC based on the flower structure shows an open circuit voltage (V_{oc}), short circuit current density (J_{sc}) and efficiency of 0.35 V, 1.82 mA and 0.28 %, respectively. The fill factor (FF) is about 0.45. The V_{oc}, J_{sc} and η of the DSSC based on nanowall structure are significantly increased about 0.531 V, 2.22 mA and 0.53 %. The increased V_{oc} of DSSC based on the nanowall structure depends on change in the energy different between Fermi band edge of the photoanode and redox potential (I^-/I^-_3) of the electrolyte [29]. The J_{sc} of the nanowall photoanode is increased from 1.82 to 2.22 mA, which is due to the increased surface area of

the film for more dye loading. The increased dye loading can be increased the light absorption which favoures to inject the more number of electrons from the excited dye molecules to conduction band of the ZnO.

Table 2. Measured and calculated parameters of the DSSC based on flower and nanowall structure.

Samples	V_{oc} (mV)	J_{sc} (mA cm^{-2})	FF (%)	η (%)
a	350	1.82	45	0.28
b	530	2.22	45	0.53

IV. CONCLUSION

ZnO and ZnO/EB thin films were electrochemically synthesized on the FTO substrate. The structural, morphological and optical properties of the thin films were successfully studied by means of XRD, FE-SEM, micro Raman and PL analysis. XRD results showed that the ZnO/EB film had a better crystallinity than the ZnO film with hexagonal wurtzite structure. FE-SEM displayed that the ZnO and ZnO/EB possessed the flower consisting of sheets, and the nanowall with porous structure, respectively. Micro Raman and PL studies indicated that the nanowall structured film had a good crystal quality with less atomic defects compared with the flower structured film. J-V results revealed that the DSSC based on nanowall structure exhibited a higher V_{oc}, J_{sc} and η than the DSSC based on flower structure. From these two DSSCs, the nanowall structured thin film is the best candidate for DSSC applications.

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